

Institute of Actuaries of India

Subject SP7 – General Insurance Reserving and Capital Modeling

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) Points to consider while choosing claim development periods in reserving:

- Claims volume –claims with low volume are unstable; this is exacerbated for quarterly claims
- Stability of data (including currency effects, if any) - if unstable then quarterly projections will be more difficult and may lead to spurious level of accuracy.
- Class of business which will impact stability of claim development pattern
- Seasonality (e.g. weather related, behavior related) - projecting by accident quarter will enable analysis of trends in quarter year on year and allow for higher development or loss ratios in some quarters
- Existence of trends
- System capability
- Availability of data
- Length of tail of data/reporting delays
- Maturity of class of business/portfolio
- Reason for review (including importance), frequency of review and prior work done
- Regulatory reporting considerations
- Management requirement – frequency of reviews i.e. quarterly, annual
- It will be easier to monitor emerging experience each quarter (Avs E) against budget if we project quarterly triangles.
- Availability of corresponding premium information
- Year end data may be more accurate as more extensively audited, quarterly data may be misleading if unaudited
- Other practical constraints such as flexibility of software, time available.
- May consider using quarterly cohorts for more recent periods and annual projection for older periods.

[0.5 each, Max 7]

ii) Recommend Mack method

- This is an easier method for the actuary to start with as it requires limited judgment.
- The formulae required for deriving the Mack standard errors are quite straightforward to implement in a spreadsheet.
- With Mack, a tail factor can be incorporated as a deterministic multiple whereas bootstrapping does not make allowance for a tail factor without adjustment
- The Mack model can handle negative incremental incurred claims movements whereas bootstrapping the ODP does not without some adjustment.

Recommend bootstrapping ODP

- This makes it easier to produce percentiles in order to produce a range
- Bootstrapping can be adjusted to make allowance for a tail factor
- Bootstrapping the ODP does not immediately allow for negative incremental incurred claims but can be adjusted to do so.
- Alternatively, bootstrapping the ODP could be applied to paid data to overcome the problems with negative incremental incurred.

[4]

iii)

a)

Incremental triangle	0	1	2	3	4	5	6
2014		10,000	20,000	15,000			
2015		15,000	15,000	100,000			
2016		20,000	15,000	25,000	14,500		
2017	20,000	10,000	15,000	20,000			
2018		15,000	25,000				
2019	20,000	20,000					
2020	2,500						

Cumulative triangle	0	1	2	3	4	5	6
2014	-	10,000	30,000	45,000	45,000	45,000	45,000
2015	-	15,000	30,000	130,000	130,000	130,000	
2016	-	20,000	35,000	60,000	74,500		
2017	20,000	30,000	45,000	65,000			
2018	-	15,000	40,000				
2019	20,000	40,000					
2020	2,500						

Incurred dev factors	3.250	2.000	1.545	1.138	1.000	1.000
Cumulative dev factors	11.428	3.516	1.758	1.138	1.000	1.000
% developed	8.8%	28.4%	56.9%	87.9%	100.0%	100.0%

UW year	Latest	Premium	% developed	BCL Ultimate	ILR	ELR Ult	BF Ult	Final Ult	Final LR
2014	45,000	60,000	100.0%	45,000	75%	45,000	45,000	45,000	75.0%
2015	130,000	75,000	100.0%	130,000	75%	56,250	130,000	130,000	173.3%
2016	74,500	100,000	100.0%	74,500	75%	75,000	74,500	74,500	74.5%
2017	65,000	110,000	87.9%	73,970	75%	82,500	75,004	73,970	67.2%
2018	40,000	90,000	56.9%	70,328	75%	67,500	69,109	70,328	78.1%
2019	40,000	120,000	28.4%	140,657	75%	90,000	104,406	104,406	87.0%
2020	2,500	125,000	8.8%	28,571	75%	93,750	88,047	88,047	70.4%

BF Ult = %developed x BCL Ult + (1-%developed) x ELR Ult.

[11]

b) Assumptions:

- Future experience is expected to be similar to past experience.
- No other claims expected in 2020 for 2020 origin year.
- For 2020, assume that the paid amount will be the same as the amount claimed (reported)
- Assume that origin years 2014-2016 are fully developed

Rationale:

1. Rationale for BF for 2020: since it is too early in the development period, BCL might under/overestimate the ultimate cost
2. Rationale for exclusion: since most of the claims seems to be around 10-20K, the claim of 100,000 seems to be an exception (or even a data error). Hence, corresponding development factors have been excluded.

[2]

c) The average of the estimated UW year LR for 2014-2020, excluding 2015 is 75.4% (depending on the period chosen for averaging)

As such, it may be argued that ILR estimate was reasonable.

[2]

[26 Marks]

Solution 2:**i) Advantages of stochastic approach:**

- Explore scenarios that might not be captured by deterministic models
 - [Appropriate example] for instance, impact of a global recession in business in various geographies.
 - Impact of correlation/diversification
- Stochastic model makes it easier to explore ripple effects, i.e. knock-on consequences.
 - [Appropriate example] for instance, major CAT event in USA triggering huge losses contributing to Underwriting CAT risk and also leading to Credit risk through reinsurer default.
- Apply different risk measures to the same set of final output. Important in sharing results with stakeholders who have different risk appetites or concerns
 - [Appropriate example] for instance, different VaR requirements for Solvency 2 and credit rating agencies (or ICA, other regulatory regimes, etc)
- Even if we have calibrated individual stresses to represent a certain degree of extremity in their individual values, it is very difficult to combine those stresses and be sure that the final result represents a sufficiently extreme combination of circumstances. A stochastic approach – in theory at least – explores all possible combinations and can rank these against the chosen risk measure
 - [Appropriate example] for instance, earthquake in Chile, floods in UK and large losses in Employer's Liability in China due to regulatory change might happen in different scenarios and can be ranked, or they might all happen in a single scenario in an extreme scenario driving the capital requirements.
(any other diversification examples)
- We can derive a probability distribution from the outcomes of a stochastic model and calculate confidence levels, if required. While the assumptions are subject to parameter error, the model is at least explicit about the assumptions being made. We can also test the assumptions using different techniques

[5]

ii) Defining UW risk

- The capital requirement for the underwriting risk is the difference between the underwriting result at the firm's chosen level of risk tolerance for the business written/earned during the modelled period, and the underwriting result on a realistic basis (or business plan).
- One way to apply a realistic basis would be to exclude any profit expected (deducting any such baseline profit from the capital requirement as a separate item).
- It would then be based on projected eventual results, that is, with best estimates of the ultimate cost of claims.

Class structure:

- Divide the firm's business into Classes based on:
 - currencies
 - territories (countries / continents)
 - LoBs (motor / property / EL / PL / BI / etc)
 - Others as per management requirements
- [Homogeneity vs data credibility]: The class structure should be decided such that there is sufficient granularity that we can consider distinctive features of the class, but not so fine that statistical methods become invalid (because of insufficient data in the subdivisions).
- We should then assess the variability of its claims and expenses, either by fitting statistical distributions or by simpler approaches such as stress tests

Claim types:

- should model **large claims separately from attritional claims** so that we can determine reinsurance recoveries directly.
- The dividing line between large and attritional claims is often the firm's typical retention for policies in the class
- Retention could be based on XoL RI covers – can vary by region / class.
- However, for classes that are small or not subject to large claims it may be more practical to model loss ratios rather than separately model individual large claims.
- We should model catastrophe-type claims separately from either, especially for events that may impact more than one class.
- Finally, we may need to consider future latent claims as a separate risk.

Large claims:

- Model large claims on a **frequency-severity** basis.
- The **Poisson and Negative Binomial** distributions could be used for frequency.
- The Poisson distribution is only appropriate where the claims are independent, since if there is any correlation between claims, this distribution will underestimate the tail risk.
- For severity, sampling from revalued past claim sizes is sometimes used, but this omits the risk of a claim greater than experienced, so it is preferable to fit a distribution.
- A heavily-skewed distribution such as the Pareto or GPDA would normally be appropriate for severity, and we should derive it from or test it against historical data revalued to current claims costs

Attritional claims:

- We generally model attritional claims in aggregate.
- A mildly-skewed distribution such as the **lognormal** may be appropriate, although again we should test this against experience.
- If the standard deviation is a sufficiently small fraction of the mean, a normal distribution may be an adequate approximation

ENIDs (or Shocks):

- For both attritional and large claims, it will be necessary to consider if there are any types of claims that are not present in the historical claims data used for parameterising the distributions. These are often referred to as Events Not In Data (ENID) or, less commonly, binary events.
- It may be appropriate to increase the standard deviation of the distributions beyond that derived purely from historical data to allow for ENIDs.
- ENIDs could also be modelled separately – this would be a more sophisticated approach, but the materiality of the ENIDs should be borne in mind.

Catastrophe:

- We often cannot model catastrophe events from the firm's experience because of their rarity
- For natural catastrophes, a proprietary model can apply a set of simulated events to the firm's exposure to derive a distribution of possible costs. It is the firm's responsibility to ensure that the model is suitable; for example, by allowing for demand surge, climate cycle, and so on
- Test the results against the known impact of recent actual catastrophes and to resolve or adjust for any discrepancy
- Human-made catastrophes such as terrorism, war, etc are generally excluded from the scope of the policy, but the insurer can develop a bespoke model if required.
- For example, for the effect of a severe recession on its creditor business, we might assess the impacts of recessions of various depths, and then model the drivers or indicators of recession to fit a distribution to these costs

Aggregate the claim distributions to give a single claim distribution for each class.

Correlations:

- Implicit correlation. examples:
 - Price inflation
 - Catastrophe events
 - Linking assumptions.
 - An example would be to express a claims inflation assumption as a margin above price inflation. This is often referred to as a 'driver' correlation

Copulas:

- Gumbel copula (stronger correlation at tail) suitable for claims distributions.
- Correlation matrix

The company may also choose to apply an explicit correlation factor (or a correlation matrix for multiple parameters).

[15]

iii)

- Commercial Property Insurance to protect against fire, earthquake and other hazards.
- Contents insurance to protect against theft of stock, flood and other hazards.
- Goods in transit to protect against loss of or damage to groceries / household items.
- BI cover to protect against loss of profits after damage to contents or warehouses.
- Motor fleet cover for accidental damage to delivery vehicles
- Motor TPL to cover liability arising from Motor accidents
- EL to protect against claim from staff
- D&O liability to protect against liability arising from wrongful doing of Directors (or officer)
- PA for staff in case they suffer certain injuries at work
- Public liability against claims from public for claims not covered under other insurance.
- Product Liability against claims from public in respect of damage / injury caused by spoilt/harmful groceries.
- Environment liability against claims for unintentional pollution
- Fidelity guarantee for fraudulent behaviour of employees, eg. theft during delivery, etc.
- Trade credit to protect against people ordering groceries and not paying for them.
- Creditor insurance to cover any loans
- Legal expenses cover to protect against legal expenses arising from complaints, food poisoning etc.
- Cyber Insurance to cover damage/liability claims as a result of hacking, cyber theft etc.
- Extended Warranty for household appliances being sold.

[8]

[28 Marks]

Solution 3:

i) The COR would not help investors to assess profitability. This is where the ROE comes in. **Limitations of the COR**

- COR does not consider all sources of return, such as the investment yield from existing capital.
- It also disregards the timing of the premiums, claims, and expenses, essentially ignoring the return on the interim investment of cash flow. The COR may not adequately describe the profitability of a business class. For instance, a COR of 115% might suggest a loss. But for a long-tail class, the cumulative return on the investment often compensates for initial losses.
- It does not recognize the relationship between classes of business or between types of insurance.

Advantages of the ROE

- In simple terms, the ROE shows the relationship between the operating result of a company and its equity.
- It can be used to compare investment in insurance companies with other types of investment, to compare insurance companies within a market, and to compare an individual class of business with others within the same company.
- Insurers can also use the ROE to determine the capacity for individual business classes and units.

[4]

ii)

a) Balance sheet as at 31st March 2020 (Amounts in Rs crores)

Company	A	B
Assets		
Total investments	12.8	355
Current Assets	0.5	8
Deferred Acquisition Costs	0.75	15
Total Assets	14.05	378
Liabilities		
O/S claims reserves	3	85
Additional URR	1.8	15
UPR	2.5	100
Current Liabilities	1.1	10
Free Reserves	5.65	168
Total Liabilities	14.05	378

Assumptions:

- Risks written uniformly during the year
- Risk is uniform across the policy year
- Only annual business underwritten
- No reinsurance

[5]

b) Assumptions

- Assume GWP = GEP
- assume AURR as at 31/03/2020 = AURR as at 31/03/2019
- assume outstanding claims reserves include IBNR

Underwriting Ratio = Loss Ratio + Expense Ratio

Loss Ratio = claims incurred/GEP

Company A = $(3.5 + 3.0 - 2.0)/5.0 = 90\%$ Company B = $(70 + 85 - 80)/200 = 37.5\%$

Expense Ratio = Acquisition Expense Ratio + Non Acquisition Expenses/GWP

Company A = $30\% + 0.5/5 = 40\%$ Company B = $15\% + 25/200 = 27.5\%$

Underwriting Ratio = Loss Ratio + Expense Ratio

A = $90\% + 40\% = 130\%$

Company B = $37.5\% + 27.5\% = 65\%$

For solvency ratio:

Solvency Ratio = Free Reserves/GWP

Company A = $5.65/5.0 = 113\%$ Company B = $168/200 = 84\%$

For return on capital employed:

Return on Capital employed = (Earned Premium – Claims Incurred – Expenses + Investment Income) / Free Reserves

Company A = $(5.0 - (3.5 + 3.0 - 20) - .5 - 1.5 + .3) / 5.65 = -21\%$

Company B = $(200 - (70 + 85 - 80) - 25 - 30 + 10) / 168.0 = 48\%$

Comments:

- Company A may have suffered from adverse claims experience due to its higher loss ratio compared to Company B.
- Both companies may be writing different classes or mix of business, each at a different point in their respective market cycle.
- Company A expense ratio is higher due to higher acquisition expense ratio.
- The company A is smaller than B and it may be spending money to expand rapidly.
- Company A solvency ratio is higher than Company B. This may be the result of a recent capital injection to expand the business.
- Company B return on capital employed is the higher, supported by a larger relative investment return compared to company A and supported by a good underwriting results.
- Relevant comment comparing profitability and solvency.

[12]

c)

- Additional Unexpired Risk Reserve carried forward / UPR carried forward
This provides a comparison of the relative profitability of the unexpired risk.
- Investment Return = Investment Income / (Current Assets + Investments)
This provides a comparison of the investment performance of the companies.
- Gross outstanding reserves / Gross written premium
This provides a comparison of relative strength of outstanding reserves
- Current Assets / Current liabilities
This provides a comparison of ability of each company to meet short term liabilities without the need to realize the investments.

[4]

[25 Marks]

Solution 4:

i) Issues:-

- Longer reporting lags to reinsurers, especially for casualty excess losses. Factors contributing to the reporting lag (time from date of accident until first notification to the reinsurer) include:
 - the cedant's perception of the claims as being reportable to the reinsurer.
 - the cedant's undervaluation of serious claims, resulting in a value below the reinsurance
 - reporting threshold (usually one half of an excess contract attachment point).
 - extreme delays in discovery/reporting of mass tort claims (e.g. asbestosis).
- Persistent upward development of most claims reserves. This may be caused by:
 - economic and social inflation.
 - a tendency of claims adjusters to reserve at the modal value.
 - a tendency to under estimate the legal costs and other direct claim handling costs
- Differing claim reporting patterns by reinsurance line, type of contract, contract terms, cedant, and intermediary (if the latter is involved).
 - This is a result of extremely heterogeneous exposures assumed by reinsurers and the low frequency with which claims are reported.
- Sparse data:

- Very few actual claims particularly for high excess non proportional business such that reserving data triangles may be very sparsely populated.
- Industry statistics are not useful, due to the heterogeneity problem. Industry loss development statistics are not directly applicable because
 - Heterogeneity of exposures, different reinsurer can experience different claims development behavior.
 - These do not apply to excess of loss coverage.
 - The severity of loss development tends to increase with the reinsurance attachment point
- Lack of information provided in reports to reinsurers.
 - For proportional covers, reinsurers often receive data reported by calendar or underwriting year, which is not useful for evaluating loss liabilities by accident year.
 - For excess covers, reinsurers often receive insufficient individual claims data.
 - In addition, premiums and losses are often reported quarterly in arrears, which adds to the IBNR exposure for both premiums and losses.
- Data coding and IT systems problems, due to heterogeneity in coverage and reporting requirements.
- Size of an adequate loss reserve is greater for a reinsurer, due to the above issues.

[5]

ii) Components of a Reinsurer's Loss Reserve:

- Case reserves (reported by the ceding company). Most pro rata contracts call for the reporting of case reserves in bulk while most excess contracts require the reporting of individual case reserve that exceed a reporting threshold amount.
- Reinsurer additional reserves on individual claims. These vary by contract and are set by the reinsurer's claims department if deemed necessary.
- Actuarial estimate of future development on (1) and (2) (Incurred but not enough reserved, IBNER).
- Actuarial estimate of "pure" IBNR. Although most actuaries would prefer separate estimates for IBNER and "pure IBNR", most reinsurer's combine these estimates and refer to them as IBNR.
- Risk load. This provision may be built in implicitly by employing conservative assumptions and loss reserving methods or may be determined explicitly. This component is extremely important for a reinsurer due to the long-tailed nature of the business assumed.

[5]

iii) Steps in Reinsurance Loss Reserving:

- Partition the reinsurance portfolio into reasonably homogeneous exposure groups. The loss reserving categories should correspond closely to the pricing categories.
- Analyze historical development patterns. Consider individual case reserve development and emergence of IBNR claims separately.
- Estimate future development. Estimate bulk reserves for IBNER and pure IBNR separately.
- Monitor and test the predictions, at least by calendar quarter.

[4]

iv)

a) Treaty Casualty excess of loss

- Long tailed business w.r.t. to claim reporting and development
- Commonly used method – chain ladder based methods, usually in conjunction with Bornhuetter-Ferguson or cape cod approach, with separate treatment of major catastrophes.

[2]

b) Treaty property catastrophe

- Shorter tail business, Relatively straightforward reserving approaches will be used, such as chain ladder or perhaps even an expected loss ratio, using rate and inflation adjusted historical loss ratios as a guide.
- Major catastrophes can be excluded and reserved separately and then can be added back at the end of the process.

[2]

v) The reinsurer has only limited control over initial underwriting and claim payments made

- Historically, some stop loss covers have been loss making.
- Direct writer's premiums generally under-priced (eg a competitive market)
- poor underwriting by direct writer
- Poor premium rating structure leading to adverse selection
- Adverse claims experience (eg large claims due to death or bodily injury or random events)
- Poor claims handling control

[3]

[21 Marks]
